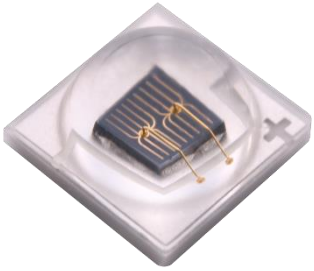
超越光电
CHAOYUE GUANGDIAN



Product Brief（产品简介）

C35C-****11C1A-LNUV

High Power Ceramic 3535 1W,3W Far IR

Description （描述）

•This surface-mount LED size is standard package: 3.45x3.45x2.26mm

•The C35C series is designed for high flux output applications with high current operation capability.

Features And Benefits（特性优点）

- Designed for high current operation
- Low thermal resistance
- Superior corrosion robustness
- Superior high efficacy at rated current enables outstanding lm/W at system level
- Viewing angle at 50%I_v : 120°
- Lumen Maintenance:Testresults according to IESNA LM-80 available
- RoHS certification

Applications （应用）

- Visual inspection
- Medical Equipment
- Automotive night vision
- Security monitoring
- Identification
- Surveillance Systems
- Scanning

Table 1. Product overview（产品总览）

Parameter	Peak Wavelength(nm)	Radiometric Power (mW)
	Rank	IF=700mA
C35C-****11CIA-LNUV	980-1005	400-700
	1005-1055	450-600
	1200-1250	100-200
	1300-1350	80-150
	1450-1550	40-100

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Performance Characteristics（性能特性）

Table 2. Product For Peak Wavelength 980nm-1550nm Electro Optical Characteristics（光电特性）, IF = 700mA ,Ta = 25℃, RH60%

Parameter	Peak Wavelength (nm)	Half Width (nm)	Voltage Rank(V)	Main Radiometric Power Rank (mW)
Model No.	Rank	Typ.	F=700mA	IF=700mA
C35C-98011C1A-LNUV	980-985	50	1.2-1.8	400-550
C35C-100011C1A-LNUV	995-1005	50	1.2-1.8	550-700
C35C-101011C1A-LNUV	1005-1015	60	1.2-1.8	450-600
C35C-103011C1A-LNUV	1025-1035	50	1.2-1.8	450-550
C35C-105011C1A-LNUV	1045-1055	50	1.2-1.8	450-550
C35C-123011C1A-LNUV	1225-1235	70	0.8-1.4	100-200
C35C-131011C1A-LNUV	1305-1315	90	0.8-1.4	80-150
C35C-132011C1A-LNUV	1315-1325	90	0.8-1.4	80-150
C35C-147011C1A-LNUV	1465-1475	110	0.6-1.2	60-100
C35C-151011C1A-LNUV	1550-1555	120	0.6-1.2	40-80

- Tolerance of measurements of the Radiometric Poweris ±7%.
- Voltage measurement tolerance is ±0.1V.
- Peak Wavelength measurement tolerance is ±5nm.
- Half Width measurement tolerance is ±5nm.
- The Radiometric Power was measured at the peak of the spatial pattern which may not be aligned with the mechanical axis of the LED package.
- The Radiometric Power table is only for reference.

Table 3. Electro and Thermal Characteristics （电热特性），IF = 700mA , Ta = 25℃, RH60%

Item	Symbol	Min	Typ	Max	Unit	Condition
Forward Voltage	VF	0.6	-	1.8	V	IF=700mA
Reverse Current	IR	-	-	10	μA	VR=5V
View Angle	2θ1/2	-	120/30	-	°	IF=700mA
Thermal Resistance	($R_{th\ j\text{-}sp}$)	-	15	-	℃/W	IF=700mA
Electrostatic Discharge	ESD	-	2000	-	V	-

• Tolerance : VF :±0.1V.
• 2Θ1/2 is the off-axis where the luminous intensity is 1/2 of the peak intensity.
• Thermal resistance : RthJS (Junction / solder) .

Table 4. Absolute Maximum Ratings （最大额定参数），Ta = 25℃, RH60%

Item	Symbol	Absolute Maximum Ratings	Unit
Forward Current	IF	1000	mA
Pulse Forward Current	IFP	1200	mA
Power Dissipation	PD	1.8	W
Reverse Voltage	VR	5	V
Operating Temperature	T_{opr}	-40 ~ +100	℃
Storage Temperature	T_{stg}	-40 ~ +100	℃
Junction Temperature	T_j	115	℃

• IFP condition with Pulse: Width≤100μs Duty cycle≤1/10.
• LED's properties might be different from suggested values like above and below tables if operation condition will be exceeded our parameter range. Care is to be taken that power dissipation does not exceed the absolute maximum rating of the product.
• All measurements were made under the standardized environment of LumiSee LED.

Product Bin（产品分Bin)

Table 5. Radiometric Power Ranks（光功率分档），Ta = 25℃, RH60%, IF=700mA

Radiometric Power (mW)			Radiometric Power (mW)		
Code	Min	Max	Code	Min	Max
P40	40	60	PC5	350	400
P60	60	80	PD0	400	450
P80	80	100	PD5	450	500
PA0	100	150	PE0	500	550
PA5	150	200	PE5	550	600
PB0	200	250	PF0	600	650
PB5	250	300	PF0	650	700
PC0	300	350			

Table 6. Forward Voltage Ranks （电压分档），Ta = 25℃, RH60%, IF=700mA

Forward Voltage Ranks(V)		
Code	Min	Max
Z06	0.6	0.8
Z08	0.8	1.0
BA0	1.0	1.2
BA2	1.2	1.4
BA4	1.4	1.6
BA6	1.6	1.8

Table 7. Peak Wavelength Ranks（波长分档）, Ta = 25℃, RH60%, IF=700mA

Peak Wavelength(nm)		
Code	Min	Max
98E	980	985
Z01	995	1005
T005	1005	1015
T025	1025	1035
T045	1045	1055
T225	1225	1235
T305	1305	1315
T315	1315	1325
T465	1465	1475
T505	1550	1555

Typical Characteristics Curve（典型特征曲线）

Fig 1. Relative Spectral Radiometric Power Distaibution for Peak Wavelength1000-1510nm, IF=700mA,Ta = 25℃, RH60%

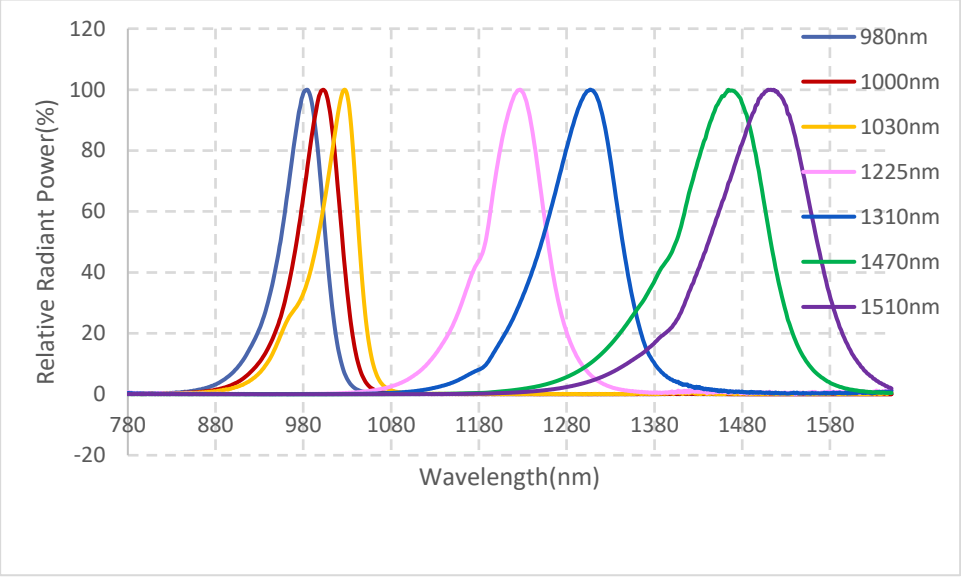


Fig 2. Viewing Angle Distribution , IF=700mA,Ta = 25℃, RH60%

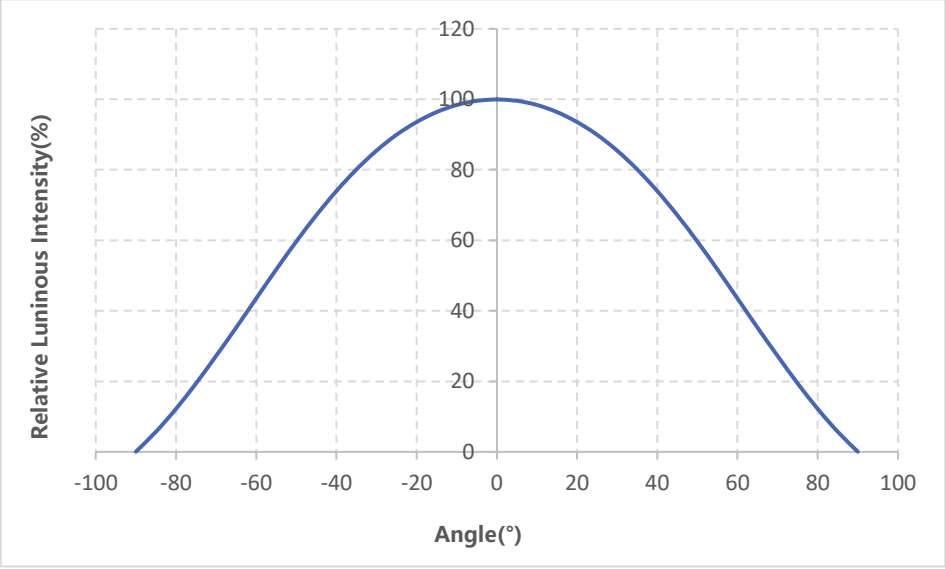


Fig 3. Relative Radiometric Power VS Forward Current,Ta = 25℃, RH60%

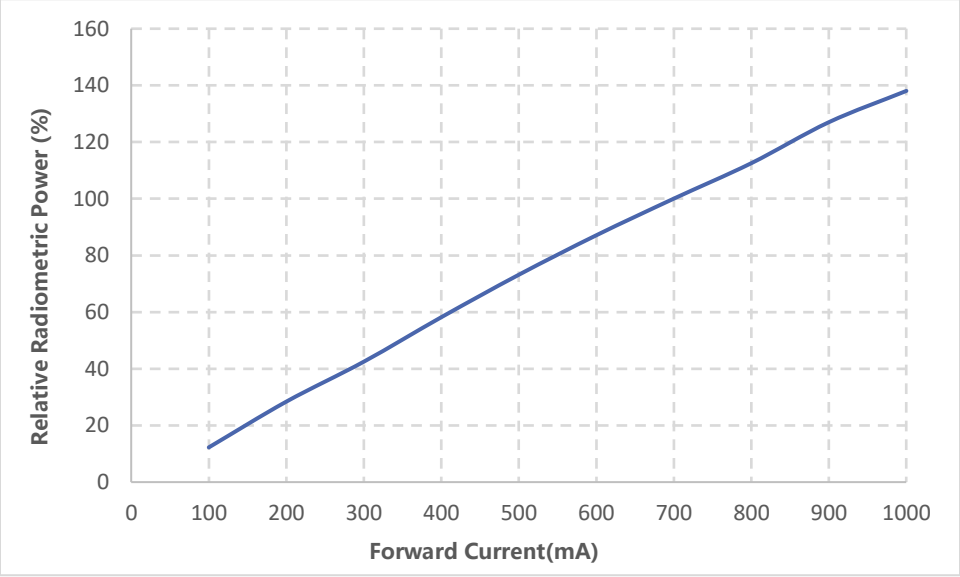


Fig 4. Forward Voltage VS. Forward Current Ta = 25℃, RH60%

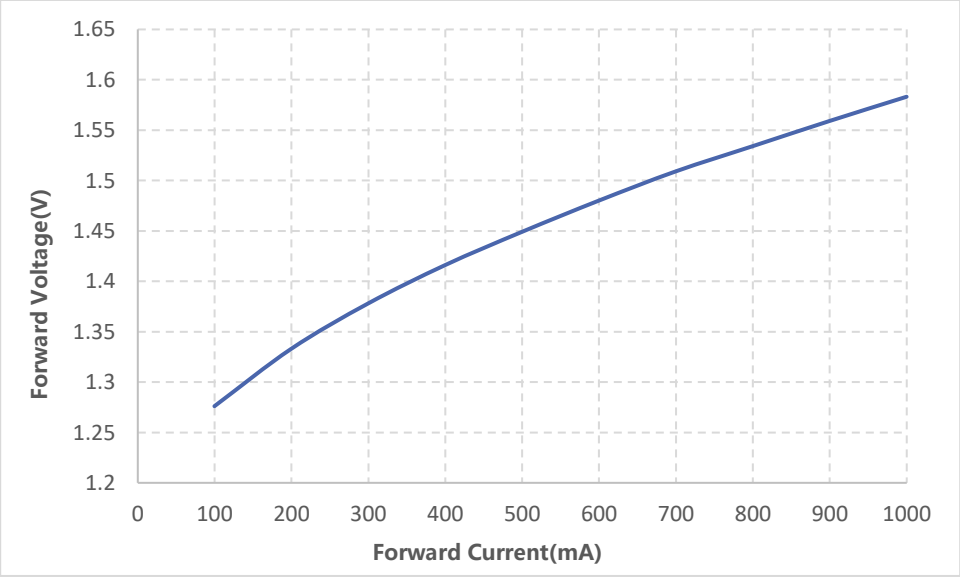


Fig 5. Relative Radiometric Power VS. Solder Temperature , IF=700mA

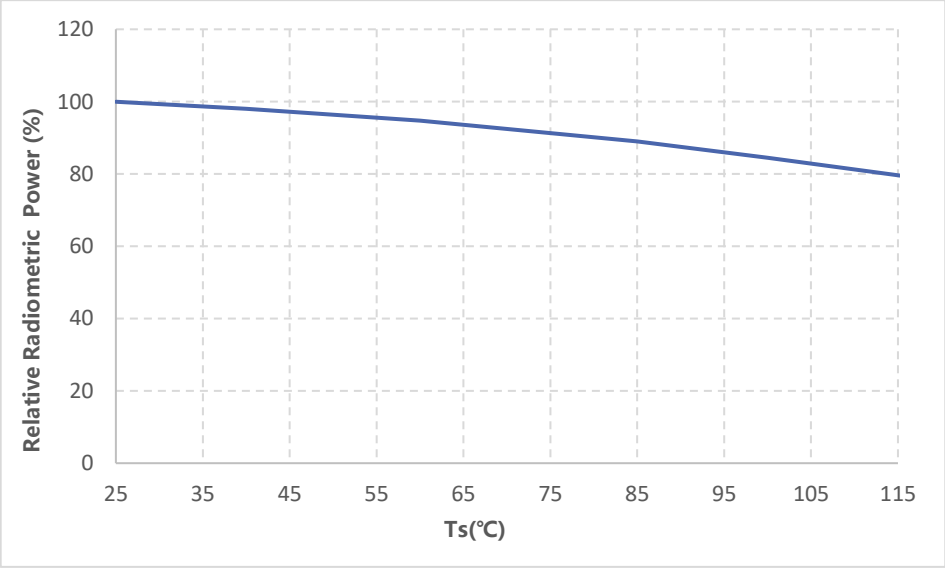
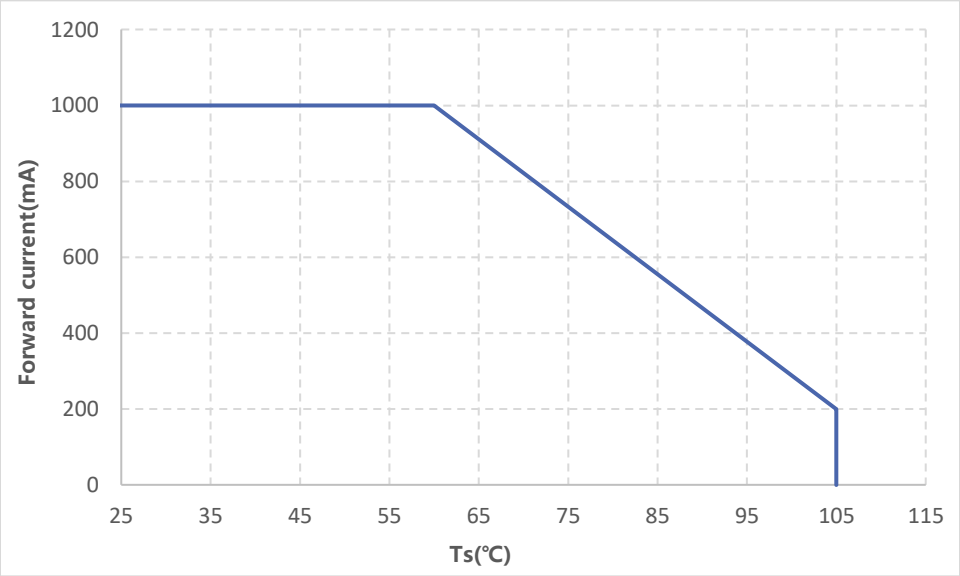


Fig 6. Forward Current VS Solder Temperature



Outline Vs. Recommended Solder Pad（外观与推荐焊盘）

Fig 7. Mechanical Dimensions（产品尺寸）

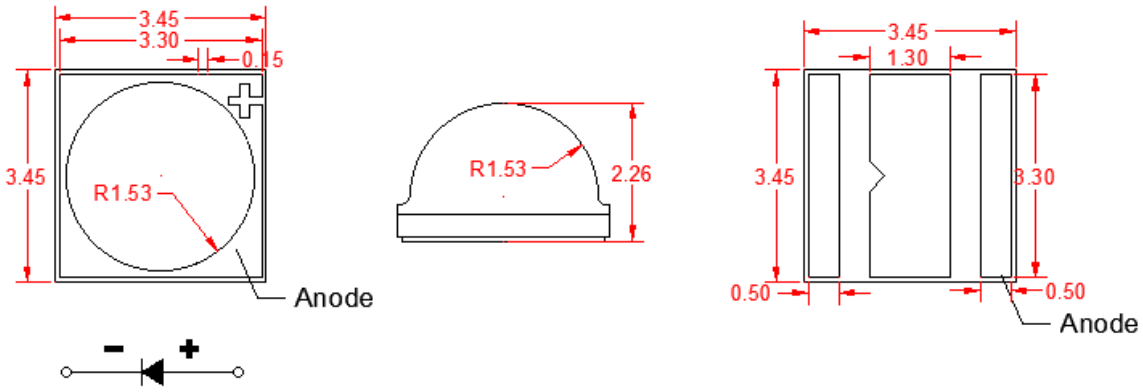
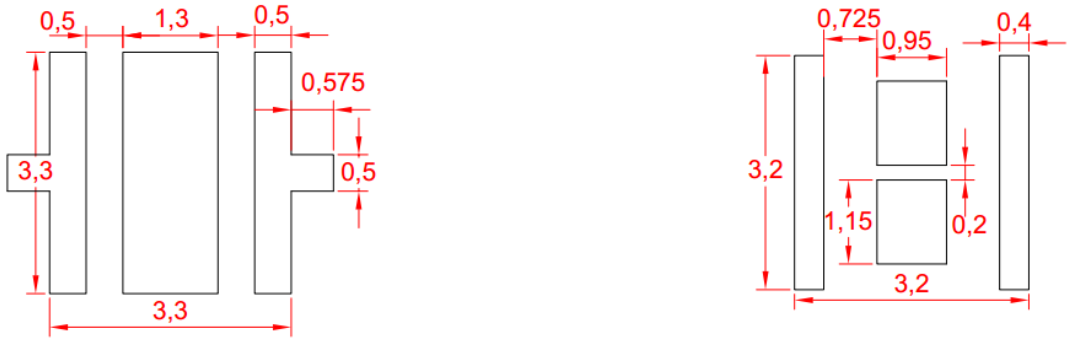


Fig 8. Recommended Solder Pad（推荐焊盘）



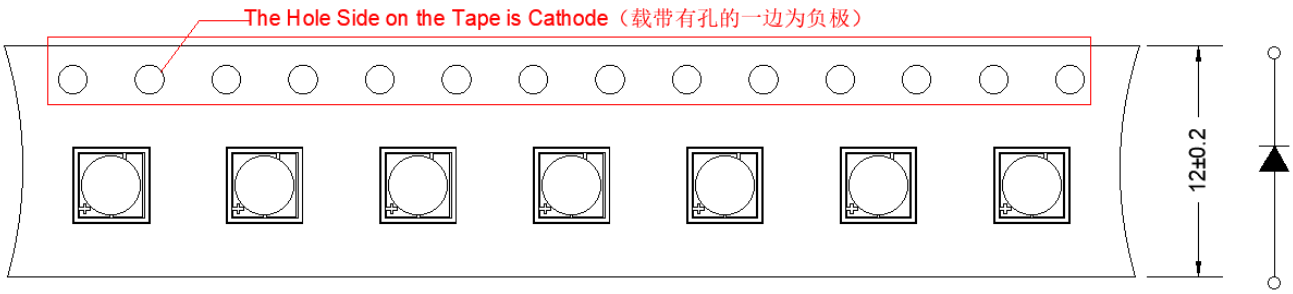
Recommended PCB Solder Pad

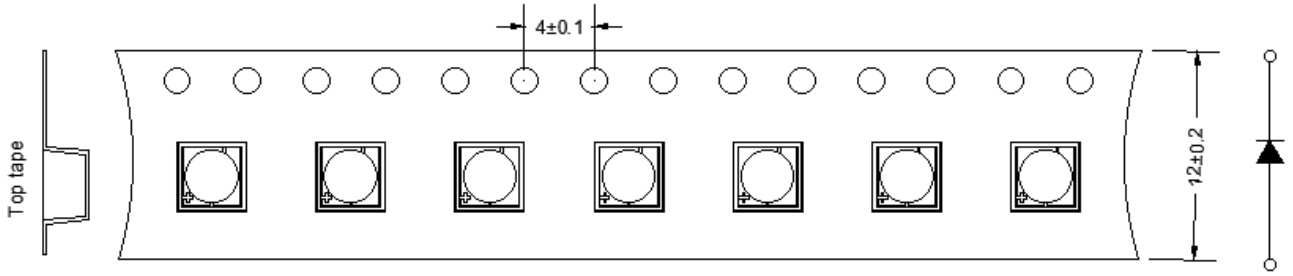
Recommended Stencil Pattern Stencil:0.12mm

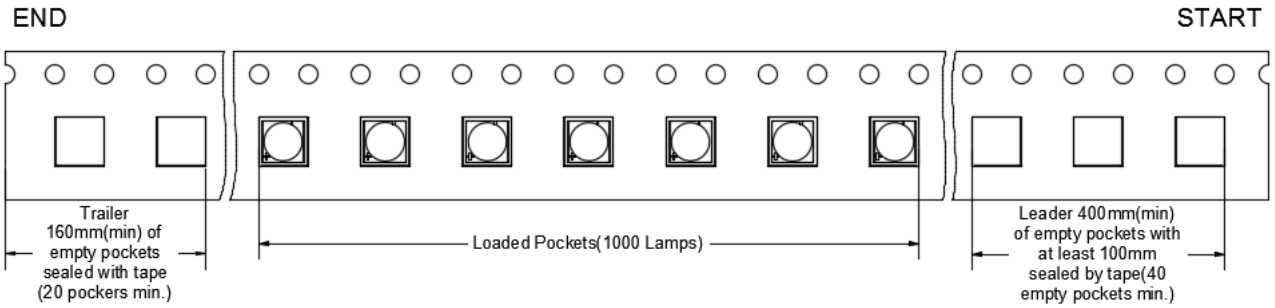
·All dimensions are in millimeters.
·Scale: none.
·Undefined tolerance is ±0.1mm.

Packaging Information（包装信息）

Fig 9. Reel Packaging 1000pcs/Reel （卷带包装, 1000pcs/卷）

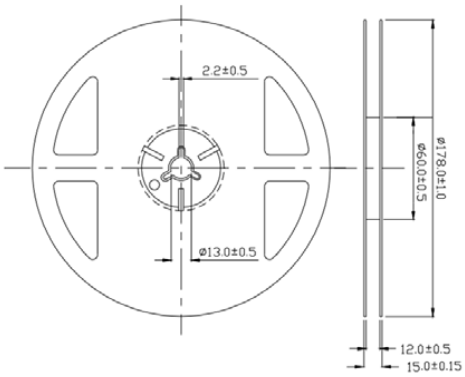


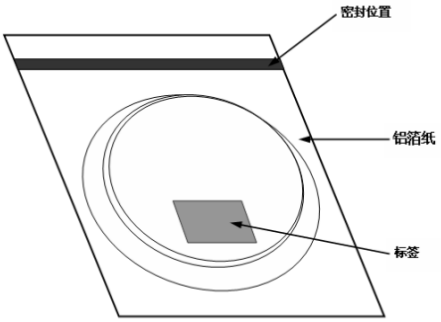




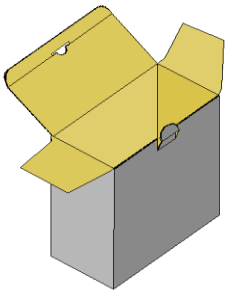
- Quantity : Max 1000pcs/Reel
- Cumulative Tolerance : Cumulative Tolerance/10 pitches to be ±0.25mm
- Adhesion Strength of Cover Tape Adhesion strength to be 0.1-0.7N when the cover tape is turned off from the carrier tape at the angle of 10˚ to the carrier tape.
- Package : P/N, Manufacturing data Code No. and Quantity to be indicated on a damp proof Package.

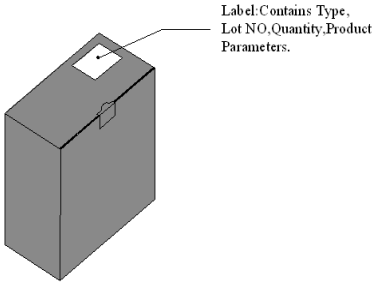
Reels（卷盘）





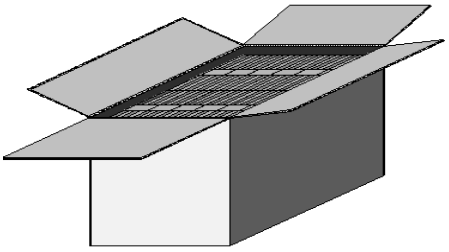
Inner Box（内箱）





•Capacity 5 or 10 reels per box（内箱容量：5或10卷）

Outer Box（外箱）



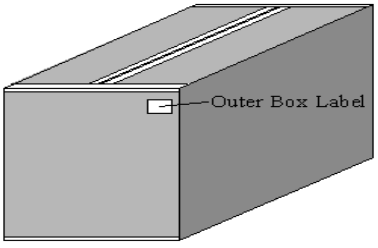


Table 8. Part Numbering System

C	□□	□	–	□□□□	□	□	□	□	-	□□□□
	X1	X2		X3	X4	X5	X6	X7		X8
C	35	C	-	1050	1	1	C1	A	-	LNUV

Item Number Code	Description	Item Number
X1	LED Type Code 产品代码	50: 5050; 30: 3030; 35: 3535;
X2	Light Color 发光颜色	W: White Color; C: Colored; I: IR; U: UV
X3	Wavelength波长	1050:1050nm
X4	No. of Serial Chip 晶片串联数量	1-Z.
X5	No. of Parallel Chip 晶片并联数量	1-Z.
X6	Lead Frame Code 支架代码	E1: EMC; E2: SMC; C1: Al_2O_3 Ceramic; C2: AlN Ceramic
X7	Viewing Angle 发光角度	A: 120 Deg. ; B: 30 Deg. ; C: 60 Deg. ; D: 90 Deg.
X8	Material Code 物料代码	LumiSee Material Code

Reflow Soldering（回流焊）

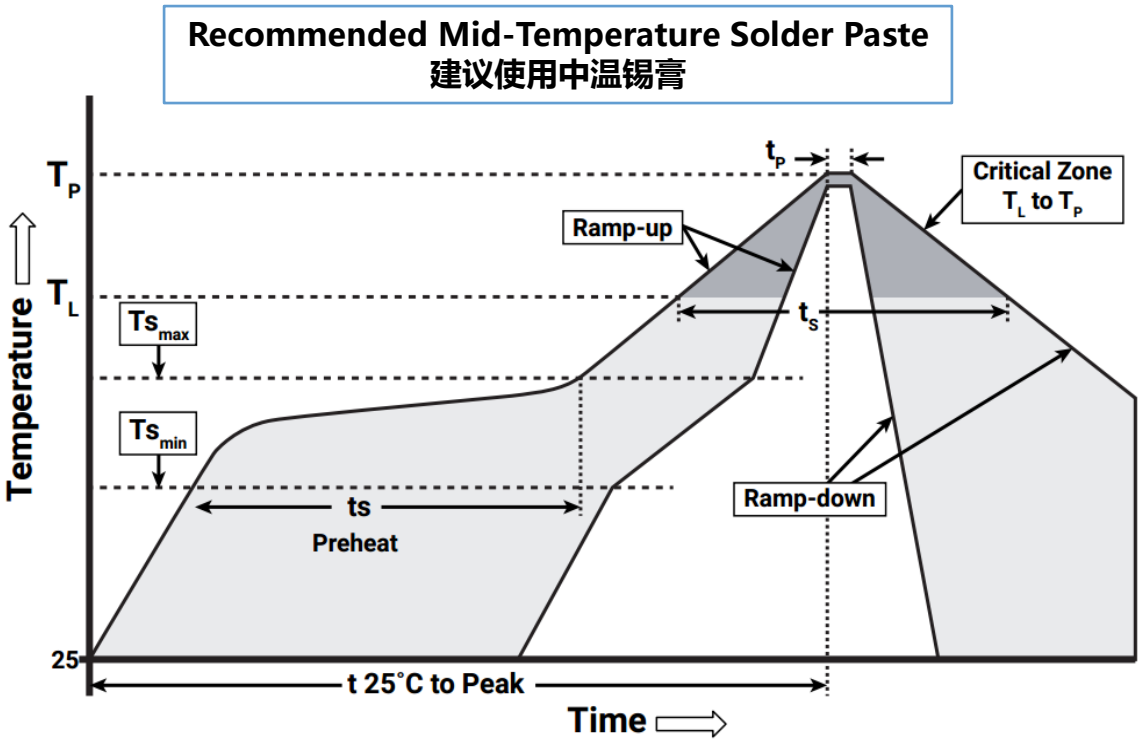


Table 9. Reflow profile characteristics

Profile Feature	Lead-Free Solder
Average Ramp-Up Rate(Ts_{max} to Tp)	1.2℃/second
Preheat:Temperature Min(Ts_{min})	120℃
Preheat:Temperature Max(Ts_{max})	170℃
Preheat:Time Max(ts_{min} to ts_{max})	65-150 seconds
Time Maintained Above:Temperature(T_L)	217℃
Time Maintained Above:Time(t_L)	45-90 seconds
Peak/Classification Temperature(Tp)	235-245℃
Time Within 5°C of Actual Peak Temperature(tp)	20-40 seconds
Ramp-Down Rate	1-6℃/ seconds
Time 25°C to Peak Temperature	4 minutes max

Pre-caution（警告）

Caution

1. Reflow soldering is recommended not to be done more than two times. In the case of more than 24 hours passed soldering after first, LED's will be damaged.
2. Repairs should not be done after the LED's have been soldered. When repair is unavoidable, suitable tools must be used.
3. Die slug is to be soldered.
4. When soldering, do not put stress on the LED's during heating.
5. After soldering, do not warp the circuit board.

Notes on LumiSee Series soldering:

1. Recommend to use reflow machine.
2. Recommend to use heating plate soldering.
3. Manual soldering is not recommended.

Notes on reflow process:

1. To confirm whether the actual temperature curve in the reflow soldering conditions comply with recommended conditions. LED's are guaranteed for one time reflow.
2. During reflow process do not apply force on LED active area.
3. After reflow process, PCB board should be cooled down before packing or storage.

Published by（出版）

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Company Profile

Set up as a high-tech company in 2011, LumiSee has been dedicated in the R&D and manufacturing of High Power Ceramic products such as 3535/5050 RGB/RGBW/R/G/B/Y LED diode, UVA and IR recognition.

Based on the technological innovation & healthy life, LumiSee sticks to safety and health, quality and innovation with the concept of creating a new vision of LED technology and providing new applications in LED industry, further more LumiSee would take good advantages of Outdoor Lighting, Plant Glowing, health and safety, energy saving and environmental protection.

With years of engineers and a professional management team in LED industry, LumiSee has established strategical cooperation with famous companies both at home and abroad, but also developed together the LED applications of curing, health, medical care, security and safety etc.

LumiSee focuses on independent innovation and R&D. Now with dozens of inventions and utility model patents having been authorized, LumiSee would continue to recruit elites for industry innovations, improvements and services. LumiSee could strive to be one of the most influential companies in the field of health and safety of global LED industry.